

$(G, \cdot)$ group if $\rightarrow$ associative
 $\rightarrow$ has an identity elem.
 $\rightarrow$ all elements have an inverse

$(R, \circ, *)$ ring if $\rightarrow (R, \circ)$ group
 $\rightarrow (R, *)$ semigroup (assoc.)
 $\rightarrow$ distributivity holds

$H \subseteq G$, $(G, \circ)$ group, then $(H, \circ)$ subgroup if: 1) H stable subset
 2) $(H, \circ)$ group.

Proving that H is a subgroup with the theorem:

$$\star \begin{cases} H \neq \emptyset, \text{id. elem.} \in H \\ (\forall) x, y \in H : x \circ y \in H \\ (\forall) x \in H : \exists x' \in H \end{cases} \quad \text{or} \quad \begin{cases} H \neq \emptyset \\ (\forall) x, y \in H : x \circ y' \in H. \end{cases}$$

Proving that S is a subring with the theorem:

$S \subseteq R$, $(R, \circ, *)$ ring, then $(S, \circ, *)$ subring if:

$$\begin{cases} S \neq \emptyset, \text{id. elem. of } " \circ " \in S. \\ (\forall) x, y \in S, x \circ y' \in S \\ (\forall) x, y, x * y \in S \end{cases}$$

$$f: (G_1, \circ) \rightarrow (G_2, *)$$

Homomorphism: $(\forall) x, y \in G_1, f(x \circ y) = f(x) * f(y)$

Isomorphism: 1) f homomorphism
 2) f bijective $\Leftrightarrow f$ invertible ($\exists f^{-1}$)

$$\textcircled{1} S_M = \{ f: M \rightarrow M \mid f \text{ bij.} \}$$

$$M \neq \emptyset$$

$(S_M, \circ)$ group?

1) associativity: $\circ$ is associative (we know) (1)

2) id. elem: $f(x) = x = 1(x) \rightarrow$ identity function (2)
 $\downarrow$
 it is bijective $\Rightarrow \in S_n$.

3) symm. element: (1) f bij. $\Rightarrow \exists f^{-1}$ inv. of f . (3)
 f^{-1} bij. $\Rightarrow f^{-1} \in S_n$.

(1), (2), (3) $\Rightarrow (S_n, \circ)$ group.

③ $H = \{z \in \mathbb{C} \mid |z| = 1\}$

a) H is a subgroup of $(\mathbb{C}^*, \cdot)$

b) H is not a subgroup of $(\mathbb{C}^*, +)$

a) $\left\{ \begin{array}{l} H \neq \emptyset \quad (1 \in H, |1| = 1) \quad (1) \\ \text{stable subset} \\ (\forall) x \in H, \exists x' \in H. \end{array} \right.$

(1) $x, y \in H \Rightarrow |x| = |y| = 1$.

$|x \cdot y| = |x| \cdot |y| = 1 \cdot 1 = 1 \Rightarrow x \cdot y \in H. \quad (2)$

(1) $x \in H, \left| \frac{1}{x} \right| = \frac{1}{1} = 1 \Rightarrow x^{-1} \in H. \quad (3)$

$$\downarrow$$

$$x^{-1} = \frac{1}{x}$$

(1), (2), (3) $\Rightarrow H \leq (\mathbb{C}^*, \cdot)$

b) give a counterexample:

$\exists 1, i \in H, |1| = |i| = 1$.

$|1+i| = \sqrt{1^2+1^2} = \sqrt{2} \neq 1 \Rightarrow 1+i \notin H$.

~~$$|x+y| = |x| + |y|$$~~

④ $U_m = \{z \in \mathbb{C} \mid z^m = 1\}, m \in \mathbb{N}^*$.

U_m is a subgroup of $(\mathbb{C}^*, \cdot)$?

$1^m = 1 \in U_m \Rightarrow H \neq \emptyset \quad (1)$

$$(v) z_1, z_2 \in U_m \Rightarrow z_1^m = 1 \Rightarrow z_1 \cdot z_2^{-1} = z_1 \cdot \frac{1}{z_2} \in U_m$$

$z_2^m = 1$ what if $z_2 = 0$?

$$(z_1 \cdot z_2^{-1})^m = \left(\frac{z_1}{z_2}\right)^m = \frac{z_1^m}{z_2^m} = \frac{1}{1} = 1 \Rightarrow z_1 \cdot z_2^{-1} \in U_m. \quad (2)$$

$$(1), (2) \Rightarrow U_m \leq (\mathbb{Q}^*, \cdot)$$

$$(5) \quad m \in \mathbb{N}, m \geq 2$$

a) $GL_m(\mathbb{C}) = \{A \in M_m(\mathbb{C}) \mid \det A \neq 0\}$ is a stable subset of $(M_m(\mathbb{C}), \cdot)$?

↳ "the general linear group of order m ".

$$(v) A, B \in GL_m, \quad \det(A \cdot B) = \underbrace{\det(A)}_{\neq 0} \cdot \underbrace{\det(B)}_{\neq 0} \neq 0 \Rightarrow A \cdot B \in GL_m.$$

$$\downarrow$$

$$\det A \neq 0$$

$$\det B \neq 0$$

~~$$\det(A+B) = \det A + \det B$$~~

b) $(GL_m(\mathbb{C}), \cdot)$ group?

" $\cdot$ " of matrices is associative (1)

I_m the id. element (matrix) $\in GL_m(\mathbb{C})$

$$\det I_m = 1 \neq 0 \Rightarrow I_m \in GL_m(\mathbb{C}) \quad (2)$$

$$(v) A \in GL_m(\mathbb{C}), \det A \neq 0 \Rightarrow \exists A^{-1} \in GL_m$$

$$\boxed{A \cdot A^{-1} = A^{-1} \cdot A = I_m}$$

$$\det(A \cdot A^{-1}) = \det(A^{-1} \cdot A) = \det I_m.$$

$$\Rightarrow \underbrace{\det A}_{\neq 0} \cdot \det(A^{-1}) = \det(A^{-1}) \cdot \underbrace{\det A}_{\neq 0} = \underbrace{1}_{\neq 0}$$

$$\Rightarrow \det(A^{-1}) \text{ must be } \neq 0. \Rightarrow A^{-1} \in GL_m(\mathbb{C}) \quad (3)$$

$$(1), (2), (3) \Rightarrow (GL_m(\mathbb{C}), \cdot) \text{ group.}$$

c) $SL_m(\mathbb{C}) = \{A \in M_m(\mathbb{C}) \mid \det A = 1\}$ is a subgroup of $(GL_m(\mathbb{C}), \cdot)$.

$$\left\{ \begin{array}{l} SL_m \neq \emptyset \quad (1) \end{array} \right.$$

$$\left\{ \begin{array}{l} (v) A, B \in SL_m, \quad A \cdot B \in SL_m \quad (2) \end{array} \right.$$

$$\left\{ \begin{array}{l} (v) A \in \text{SL}_m, A^{-1} \in \text{SL}_m \quad (3) \end{array} \right.$$

$$I_m \in \text{SL}_m(\mathbb{C}) \Rightarrow \text{SL}_m(\mathbb{C}) \neq \emptyset \quad (1).$$

$$(iv) A, B \in \text{SL}_m, \det A = \det B = 1.$$

$$\det(A \cdot B) = \det A \cdot \det B = 1 \cdot 1 = 1. \Rightarrow A \cdot B \in \text{SL}_m \quad (2).$$

$$(v) A \in \text{SL}_m(\mathbb{C}), \exists A^{-1} \in \text{SL}_m(\mathbb{C})$$

$$\Downarrow$$

$$\det A \neq 0$$

$$\det A^{-1} = \frac{1}{\det A} = \frac{1}{1} = 1. \Rightarrow A^{-1} \in \text{SL}_m(\mathbb{C})$$

$$\textcircled{6} \quad i) \mathbb{Z}[i] = \{a+bi \mid a, b \in \mathbb{Z}\} \text{ subring of } (\mathbb{C}, +, \cdot)?$$

$$\left\{ \begin{array}{l} \mathbb{Z}[i] \neq \emptyset, \text{ id. elem of } "+" \in \mathbb{Z}[i] \\ (v) z_1, z_2 \in \mathbb{Z}[i] : z_1 + z_2' \in \mathbb{Z}[i] \\ (vi) z_1, z_2, z_1 \cdot z_2 \in \mathbb{Z}[i] \end{array} \right.$$

$$2+3i \in \mathbb{Z}[i] \Rightarrow \mathbb{Z}[i] \neq \emptyset \quad (1)$$

$$z_1 = a_1 + b_1 i \in \mathbb{Z}[i] \Rightarrow a_1, b_1 \in \mathbb{Z}$$

$$z_2 = a_2 + b_2 i \in \mathbb{Z}[i] \Rightarrow a_2, b_2 \in \mathbb{Z}$$

$$z_1 + z_2 = a_1 + b_1 i + a_2 + b_2 i = \underbrace{a_1 + a_2}_{\in \mathbb{Z}} + \underbrace{(b_1 + b_2)}_{\in \mathbb{Z}} i \in \mathbb{Z}[i]$$

$$(v) z_2 = a_2 + b_2 i \in \mathbb{Z}[i], \exists \underbrace{-a_2}_{\in \mathbb{Z}} - \underbrace{b_2}_{\in \mathbb{Z}} i \in \mathbb{Z}[i]$$

$$z_1 + z_2' = a_1 + b_1 i - a_2 - b_2 i = \underbrace{a_1 - a_2}_{\in \mathbb{Z}} - \underbrace{(b_1 + b_2)}_{\in \mathbb{Z}} i \in \mathbb{Z}[i] \quad (2)$$

$$\begin{aligned} z_1 \cdot z_2 &= (a_1 + b_1 i)(a_2 + b_2 i) = a_1 a_2 + a_1 b_2 i + a_2 b_1 i - b_1 b_2 \\ &= \underbrace{a_1 a_2 - b_1 b_2}_{\in \mathbb{Z}} + \underbrace{(a_1 b_2 + a_2 b_1)}_{\in \mathbb{Z}} i \in \mathbb{Z}[i] \quad (3) \end{aligned}$$

$$(1), (2), (3) \Rightarrow \mathbb{Z}[i] \text{ subring.}$$

$$\textcircled{7} \quad b) g: \mathbb{C}^* \rightarrow \text{GL}_2(\mathbb{R})$$

$$g(a+bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

g homomorphism of $(\mathbb{C}^*, \cdot) \rightarrow (GL_2(\mathbb{R}), \cdot)$

$$(1) z_1 = a_1 + b_1 i \in \mathbb{C}^*$$

$$z_2 = a_2 + b_2 i \in \mathbb{C}^*$$

$$g(z_1 \cdot z_2) \stackrel{?}{=} g(z_1) \cdot g(z_2)$$

$$\Leftrightarrow g((a_1 + b_1 i)(a_2 + b_2 i)) \stackrel{?}{=} g(a_1 + b_1 i) \cdot g(a_2 + b_2 i)$$

$$\Leftrightarrow g(a_1 a_2 - b_1 b_2 + (a_1 b_2 + a_2 b_1)i) \stackrel{?}{=} \begin{pmatrix} a_1 & b_1 \\ -b_1 & a_1 \end{pmatrix} \cdot \begin{pmatrix} a_2 & b_2 \\ -b_2 & a_2 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} a_1 a_2 - b_1 b_2 & a_1 b_2 + a_2 b_1 \\ -(a_1 b_2 + a_2 b_1) & a_1 a_2 - b_1 b_2 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} a_1 a_2 - b_1 b_2 & a_1 b_2 + b_1 a_2 \\ -(a_2 b_1 + a_1 b_2) & a_1 a_2 - b_1 b_2 \end{pmatrix} \text{ adv.}$$

$\Rightarrow g$ is a homomorphism.

⑧ $m \in \mathbb{N}, m \geq 2$.

$$(\mathbb{Z}_m, +) \simeq (U_m, \cdot)$$

isomorphic.

$$\mathbb{Z}_m = \{\hat{0}, \hat{1}, \dots, \hat{m-1}\}$$

$$U_m = \{z \in \mathbb{C} \mid z^m = 1\} = \{\varepsilon^0 = 1, \varepsilon^1, \dots, \varepsilon^{m-1}\}.$$

! Two sets are isomorphic if they have the same cardinal.

$$|\mathbb{Z}_m| = |U_m| = m.$$

I need to find a function between them and prove it's isomorphism.

$$f: \mathbb{Z}_m \rightarrow U_m, f(\hat{x}) = \varepsilon^x$$

$$\Rightarrow f^{-1}(\hat{x}) = \log \varepsilon^x$$

• homomorphism

$$(1) \hat{x}, \hat{y} \in \mathbb{Z}_m : f(\hat{x} + \hat{y}) = f(\hat{x}) \cdot f(\hat{y})$$

$$f(\hat{x} + \hat{y}) = \varepsilon^{\hat{x} + \hat{y}} = \varepsilon^{\hat{x}} \cdot \varepsilon^{\hat{y}} = f(\hat{x}) \cdot f(\hat{y})$$

$\Rightarrow f$ homomorphism (1)

$$f(f^{-1}(\hat{x})) = x$$

$$\Leftrightarrow f(\log \varepsilon^{\hat{x}}) = \hat{x}$$

$$\Leftrightarrow \varepsilon^{\log \varepsilon^{\hat{x}}} = \varepsilon^{\hat{x}}$$

• bijective

$$\exists f^{-1} \text{ s.t. } (\forall) \hat{x} \in \mathbb{Z}_m: (f \circ f^{-1})(\hat{x}) = (f^{-1} \circ f)(\hat{x}) = \hat{x}$$

$$(f \circ f^{-1})(\hat{x}) = f(f^{-1}(\hat{x})) = \hat{x}$$

$$\Leftrightarrow f(\log_2 \hat{x}) = \hat{x}$$

$$\Leftrightarrow 2^{\log_2 \hat{x}} = \hat{x}$$

$$\Leftrightarrow \hat{x} = \hat{x} \quad \text{true}$$

$$\Rightarrow \exists f^{-1} \Rightarrow f \text{ bij. (2)}$$

$$(1), (2) \Rightarrow (\mathbb{Z}_m, +) \simeq (V_m, \cdot)$$

! In the exam you will need to

find the function f
prove that 2 things are isomorph..
cardinal equal !